
GlucoTwin
A metabolic & glucose control digital twin
HACKATHON DEMO · EDUCATIONAL, NOT MEDICAL ADVICE

THE SCALE OF THE PROBLEM

830M

people were living with diabetes worldwide in 2022

WHO estimate, 2022

Diabetes is a scale problem — and a personalization problem

In 2022, 59% of adults aged 30+ living with diabetes were not taking medication. Patients need a clearer, personal picture of how daily choices affect their glucose.



WHAT GLUCOTWIN IS

A digital twin to explore glucose control before acting

Metabolic & Glucose Control Digital Twin

Patient profile, scenario simulations and saved history

Educational decision support: explore projected outcomes, not medical advice.

LIVE DEMO JOURNEY

Judges reach the simulation in three steps

Start on the welcome screen, choose a path, and press Get Started.

01

Welcome

Open the welcome screen of the app.

02

Choose a path

Sign in to sync data to the cloud, or use Demo Mode.

03

Get Started

Enter the patient profile and simulation experience.



THE CORE PRODUCT LOOP

The product loop is built around continuity

Patient profile

Demo Mode includes a patient profile; signed-in users can sync it.

Simulation

Carbs, walking, sleep, protein and fat drive each projection.

History

Local session history, plus saved runs when cloud is configured.

Cloud sync

Sign in to sync profile and simulation history to the cloud.

HOW THE TWIN WORKS

Scenario inputs become a projected response and risk level

01

Inputs

Carbs, walking minutes, sleep hours, protein and fat.

02

Project

Calculates projected glucose response and peak.

03

Risk

Returns a risk level, saved with simulation history.

Compare scenarios to see how each input shifts the projection.



HISTORY AND CONTINUITY

Every run builds a history for the next decision

Demo Mode

Works without an account, with local session history.

Patient profile

Profile and simulation runs are organized per patient.

Continuous

Simulation history is kept across runs, and scenarios can be compared.

Cloud sync

Signed-in users sync profile and history to the cloud.

SHOWN VS. TO VALIDATE

What is shown today, and what still needs proof

01

Shown

Welcome, Demo Mode, simulation, history and compare-scenario flows.

02

Concept

Clinician plans, clarification requests and secure patient messaging.

03

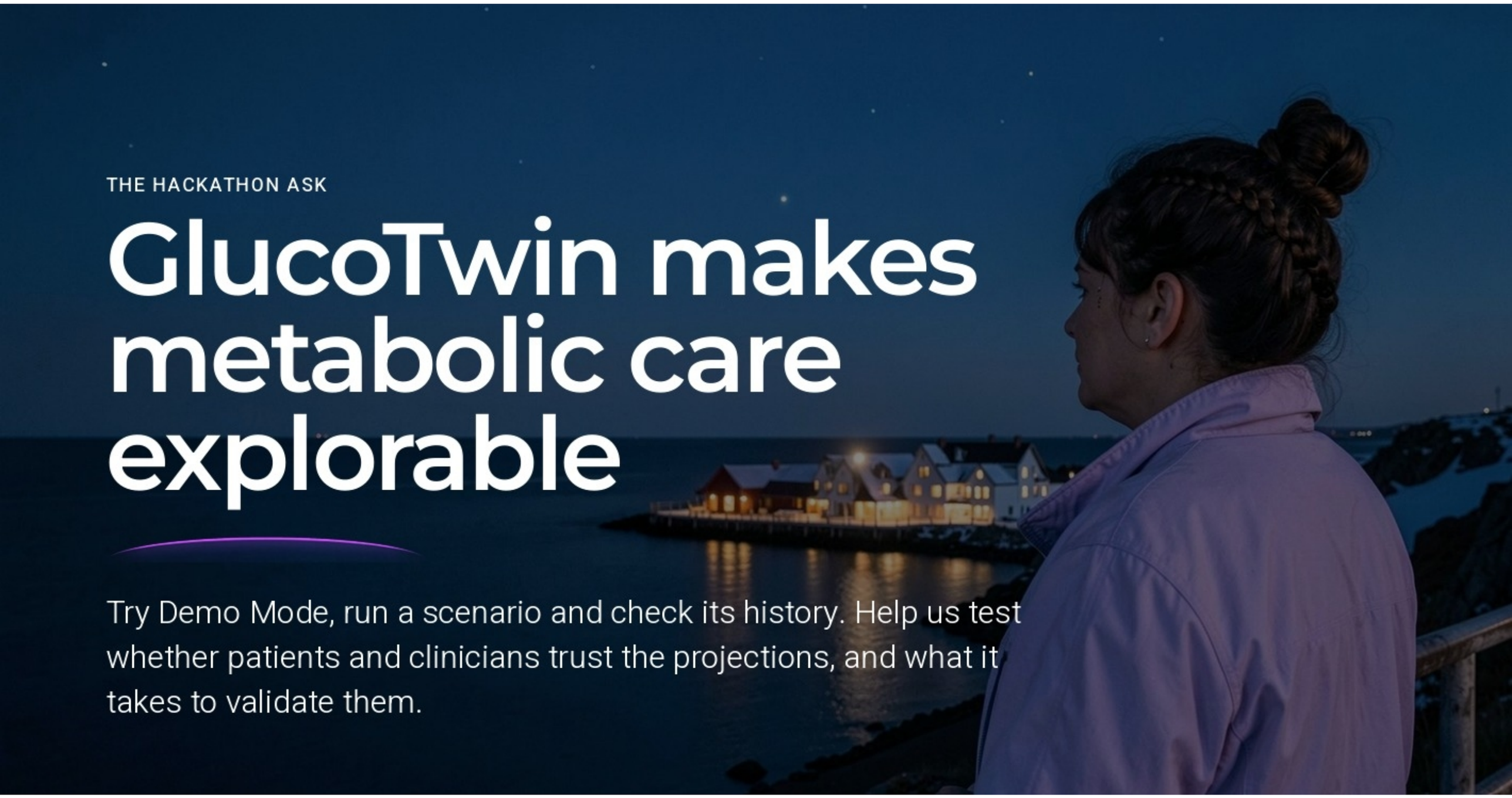
Validate

Model accuracy, clinical validation and safety limits are not yet proven.

04

Secure

Privacy, consent and audit trails are needed before real patient use.



THE HACKATHON ASK

GlucoTwin makes metabolic care explorable

Try Demo Mode, run a scenario and check its history. Help us test whether patients and clinicians trust the projections, and what it takes to validate them.